

THE THIRTY-FIRST WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 5, 1970

A-1. Show that the power series for the function

$$e^{ax} \cos bx \quad (a > 0, b > 0)$$

in powers of x has either no zero coefficients or infinitely many zero coefficients.

A-2. Consider the locus given by the real polynomial equation

$$Ax^2 + Bxy + Cy^2 + Dx^3 + Ex^2y + Fxy^2 + Gy^3 = 0,$$

where $B^2 - 4AC < 0$. Prove that there is a positive number δ such that there are no points of the locus in the punctured disk

$$0 < x^2 + y^2 < \delta^2.$$

A-3. Find the length of the longest sequence of equal nonzero digits in which an integral square can terminate (in base 10) and find the smallest square which terminates in such a sequence.

A-4. Given a sequence $\{x_n\}$, $n = 1, 2, \dots$, such that $\lim_{n \rightarrow \infty} \{x_n - x_{n-1}\} = 0$.

Prove that

$$\lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{n} = 0.$$

A-5. Determine the radius of the largest circle which can lie on the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad (a > b > c).$$

A-6. Three numbers are chosen independently at random, one from each of the three intervals $[0, L_i]$ ($i = 1, 2, 3$). If the distribution of each random number is uniform with respect to length in the interval it is chosen from, determine the expected value of the smallest of the three numbers chosen.

B-1. Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n^4} \prod_{i=1}^{2n} (n^2 + i^2)^{1/n}.$$

B-2. The time-varying temperature of a certain body is given by a polynomial in the time of degree at most three. Show that the average temperature of the body between 9 A.M. and 3 P.M. can always be found by taking the average of the temperatures at two fixed times, which are independent of which polynomial occurs. Also, show that these two times are 10:16 A.M. and 1:44 P.M. to the nearest minute.

B-3. A closed subset S of R^2 lies in $a < x < b$. Show that its projection on the y -axis is closed.

B-4. An automobile starts from rest and ends at rest, traversing a distance of one mile in one minute, along a straight road. If a governor prevents the speed of the car from exceeding ninety miles per hour, show that at some time of the traverse the acceleration or deceleration of the car was at least 6.6 ft./sec.²

B-5. Let u_n denote the "ramp" function

$$u_n(x) = \begin{cases} -n & \text{for } x \leq -n, \\ x & \text{for } -n < x \leq n, \\ n & \text{for } x > n, \end{cases}$$

and let F denote a real function of a real variable. Show that F is continuous if and only if $u_n \circ F$ is continuous for all n . (Note: $(u_n \circ F)(x) = u_n[F(x)]$.)

B-6. A quadrilateral which can be inscribed in a circle is said to be *inscribable* or *cyclic*. A quadrilateral which can be circumscribed to a circle is said to be *circumscribable*. Show that if a circumscribable quadrilateral of sides a, b, c, d has area $A = \sqrt{abcd}$, then it is also inscribable.